Project Title

Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Report Submitted ***UNIFIED MENTOR*** *in Partial fulfillment of the requirements for the degree of*

Master in Business Administration

Submitted by:
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Under the Guidance of:
UNIFIED MENTOR

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DECLARATION

I ,**SANDEEP KUMAR MISHRA**, hereby declare that the **Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence** prepared by me under the guidance of **Unified Mentor**

I also declare that this **Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence** is towards partial fulfillment of the Unified Mentor

I have undergone a the **Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence** . I further declare that this report is based on an Original study undertaken by me and has not been submitted earlier for the award of any degree or diploma from any other University/ Institution.

Place: Gurgaon, Haryana **SANDEEP KUMAR MISHRA**
Date:

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1. Executive Summary

This report presents a comprehensive, data-driven buyer segmentation and investment profiling framework developed for Parcl Co. Limited, in collaboration with Unified Mentor. Using unsupervised machine learning techniques applied to real client and property transaction data, the project uncovers hidden patterns in buyer behavior that traditional descriptive analytics cannot detect.

2,000	10,000	7,305	4
Clients Analyzed	Property Listings	Sold Transactions	Buyer Segments Identified

The analysis combined client demographic and behavioral records with property acquisition history to build a unified feature set spanning demographics, financing behavior, satisfaction, and investment value. K-Means clustering, validated against Hierarchical (Agglomerative) clustering, was used to segment the buyer base into four statistically distinct and business-relevant groups.

Key Findings

- Four distinct buyer segments were identified: **Luxury Home Buyers**, **Investment-Focused Buyers**, **First-Time / Value Buyers**, and **Corporate Buyers**.
- The optimal number of clusters (k = 4) was confirmed using both the Elbow Method and Silhouette Analysis, and cross-validated with Hierarchical Clustering (Adjusted Rand Index ≈ 0.68).
- Investment-purpose buyers represent roughly 23% of the client base but concentrate a disproportionate share of total portfolio value.
- Corporate buyers, though only about 5% of clients, show the highest average loan utilization and the youngest average buyer age.
- The USA dominates the client base across all segments, with the UK, Canada, and Germany forming the next tier of international buyer activity.

Strategic Impact

These findings enable Parcl to move from a one-size-fits-all marketing approach toward precision-targeted campaigns, tailored property recommendations, and segment-specific financing strategies — directly addressing the inefficiencies identified in the original problem statement.

2. Project Background & Problem Statement

2.1 Background

Real estate companies operate in markets where buyer behavior is highly diverse, spanning individual home buyers, institutional investors, international buyers, high-net-worth investors, and first-time buyers. Without a structured segmentation approach, companies tend to treat all buyers identically, which leads to inefficient marketing spend, poor customer targeting, and missed investment opportunities.

Parcl Co. Limited, in partnership with Unified Mentor, initiated this project to apply AI-based clustering techniques to discover hidden patterns in buyer behavior using real transactional and demographic data collected across its property portfolio.

2.2 Problem Statement

Parcl currently lacks a data-driven understanding of:

- Different types of property buyers active on the platform
- Investment motivations across demographic groups
- Geographic differences in investment behavior
- Customer financing patterns and loan dependency

Left unaddressed, this leads to:

- Inefficient marketing spend across undifferentiated buyer groups
- Generic property recommendations that do not match buyer intent
- Poor investor targeting and missed high-value opportunities

2.3 Project Objectives

- Clean and consolidate client and property transaction data into an analysis-ready dataset
- Engineer behavioral and financial features that capture buyer intent and investment value
- Apply K-Means and Hierarchical clustering to segment buyers into meaningful groups
- Validate the optimal number of segments using the Elbow Method and Silhouette Analysis
- Translate clusters into actionable buyer personas and business recommendations
- Deliver an interactive Streamlit dashboard for ongoing segment monitoring

3. Dataset Overview & Description

Two source datasets were provided and merged to build the analytical base table: a client-level demographic and behavioral file, and a property-level transaction file capturing purchase history across residential and commercial units.

3.1 Client Dataset

2,000 unique client records with the following fields:

Field	Description
client_id	Unique client identifier
client_type	Individual or Company
gender	Gender of buyer
date_of_birth	Used to derive buyer age
country / region	Geographic location of buyer
acquisition_purpose	Home use or Investment
loan_applied	Whether financing was used
referral_channel	Website, Agency, or Client referral
satisfaction_score	Customer satisfaction rating (1-5)

3.2 Property Transactions Dataset

10,000 property listing/transaction records with the following fields:

Field	Description
Listing id / tower number / unit number	Property identifiers
Transaction date	Date of sale or listing
Unit category	Apartment or Office
Floor area sqft	Unit size in square feet
Sale price	Transaction value (USD)
Listing status	Sold or Available
Client ref	Links transaction to client id

3.3 Dataset Summary Statistics

Metric	Value
Total Clients	2,000
Total Property Records	10,000
Sold Transactions	7,305 (73.1%)
Available Listings	2,695 (26.9%)
Total Transaction Value	$2,520,750,960.84
Average Sale Price	$345,072.00
Transaction Date Range	Jan 2024 – Dec 2025
Countries Represented	10

4. Data Science Methodology

A structured, six-step data science pipeline was followed, moving from raw data cleaning through to final cluster interpretation. This section documents Steps 1–3 (Data Cleaning, Feature Engineering & Encoding, and Feature Scaling); modeling steps are covered in Sections 6–8.

4.1 Data Cleaning

- **Missing values:** The client and property datasets contained no missing values in core identifying or behavioral fields; the only systematic gap was *client ref* in property records where a unit remained unsold (Available status), which was expected and retained as-is.
- **Date normalization:** *date of birth* and *transaction date* fields contained mixed date formats (MM-DD-YYYY and MM/DD/YYYY) and were parsed into a single standardized date time format.
- **Currency cleaning:** *sale price* values were stored as formatted strings (e.g. "$300,385.62") and converted to numeric floats for computation.
- **Duplicate removal:** Client records were checked and de-duplicated on *client id*; no duplicate client entries were found.
- **Categorical normalization:** Text labels (e.g. gender, client type, acquisition purpose) were checked for consistent casing and spelling across the dataset.

4.2 Feature Engineering & Encoding

Client and property data were merged at the client level, aggregating each client's sold transactions into behavioral investment features:

- **age** — derived from date of birth relative to the analysis reference date
- **total units** — count of properties purchased by the client
- **total investment** — sum of sale prices across all purchases
- **avg unit price** — mean transaction value per client
- **avg floor area** — mean unit size purchased

Categorical variables were encoded for clustering compatibility:

- **Label Encoding** applied to binary/ordinal fields: client type, gender, acquisition purpose, loan applied, referral channel
- **One-Hot Encoding** applied to country for geographic segmentation views in the dashboard layer

4.3 Feature Scaling

Because K-Means relies on Euclidean distance, all numeric features were standardized using **Standard Scaler** (zero mean, unit variance) prior to clustering. This prevents high-magnitude features such as total investment (measured in hundreds of thousands of dollars) from dominating distance calculations over smaller-scale features such as satisfaction score.

The final scaled feature matrix used for clustering included: age, satisfaction score, total units, total investment, avg unit price, avg floor area, client type, acquisition purpose, and loan applied.

5. Exploratory Data Analysis

Before modeling, exploratory analysis was performed to understand distributions, relationships, and data quality across the merged dataset. Key visualizations are presented below.

5.1 Client Demographics

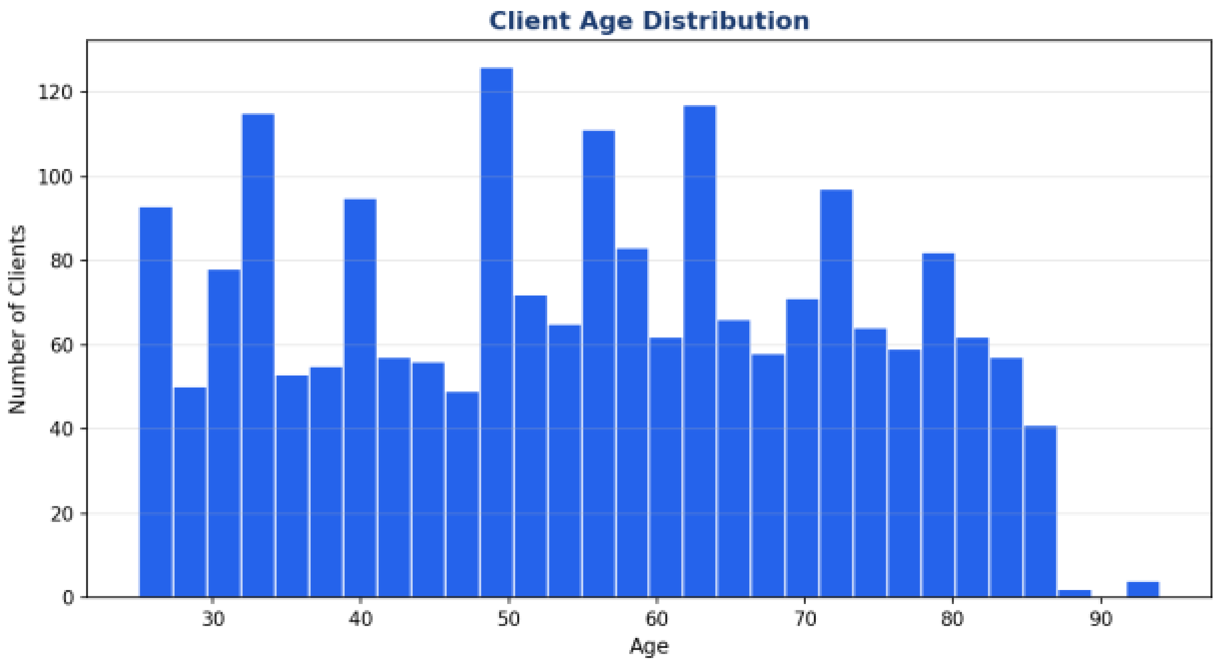

Figure 5.1 — Distribution of client age across the buyer base (mean age ≈ 55 years).

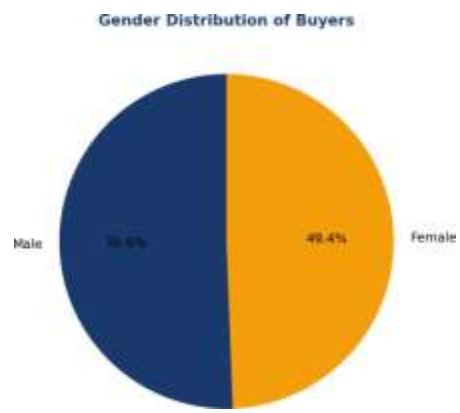

Figure 5.2 — Gender split of buyers.

5.2 Client Type & Property Mix

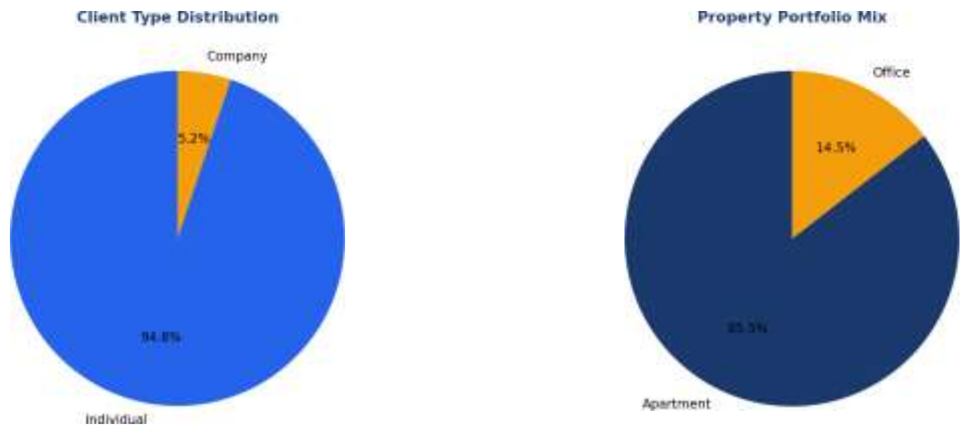

Figure 5.3 — Client type distribution (left) and property portfolio mix (right).

5.3 Geographic Distribution

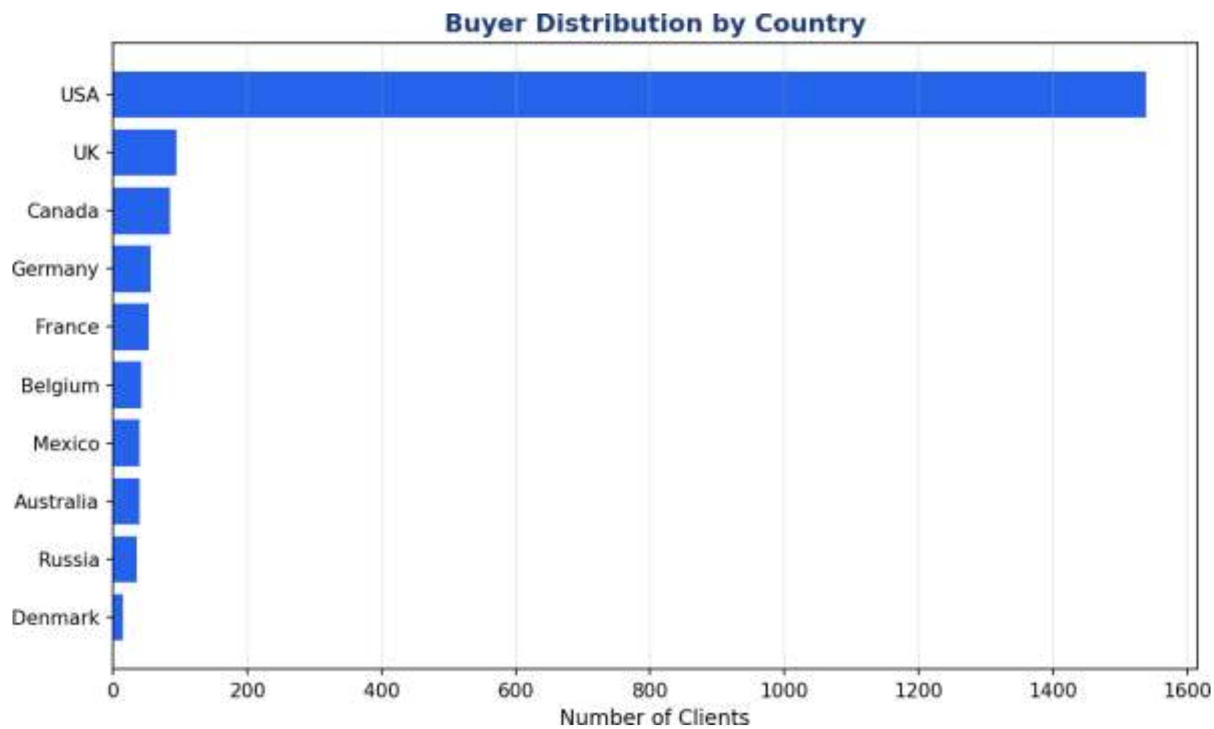

Figure 5.4 — Buyer distribution by country of residence. The USA accounts for the majority of the client base (76.9%), followed by the UK and Canada.

5.4 Acquisition Behavior

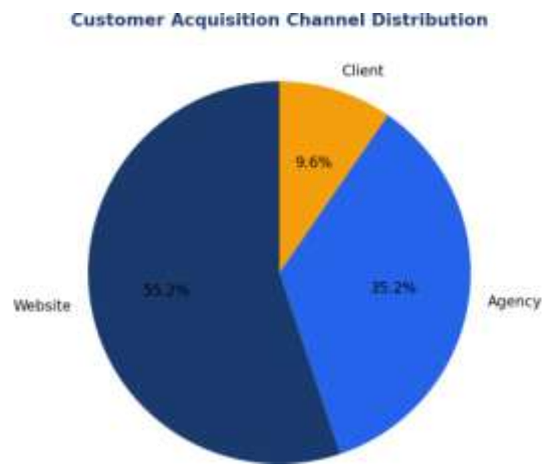

Figure 5.5 — Distribution of customer acquisition channels. Website-driven acquisition leads at 55.2%, followed by Agency referrals at 35.3%.

5.5 Key EDA Observations

- Buyer age is broadly distributed from 25 to 94 years, with a concentration in the 40–70 range, reflecting a mature, established buyer base.
- Individual buyers dominate the client base (94.9%), with corporate buyers forming a smaller but financially significant segment (5.1%).
- Apartments make up 85.5% of the property portfolio, with office units comprising the remaining 14.5%.
- The USA is the dominant market by client volume; California, Nevada, and Colorado are the leading regions.
- Satisfaction scores are roughly evenly distributed from 1 to 5, suggesting no single acquisition channel or buyer type is systematically over- or under-satisfied.

6. Clustering Model Selection

Two complementary unsupervised clustering approaches were applied to the standardized feature matrix: K-Means Clustering as the primary segmentation model, and Hierarchical (Agglomerative) Clustering as an independent validation method.

6.1 K-Means Clustering

K-Means partitions clients into k clusters by minimizing within-cluster variance (inertia). It was selected as the primary method for its computational efficiency at scale and ease of interpretation for business stakeholders.

Advantages:

- Efficient on large datasets (fast convergence on 2,000 client records)
- Produces easily interpretable, centroid-based cluster definitions
- Well suited to continuous behavioral and financial features

The model was fit using the k-means++ initialization strategy with 10 independent restarts (n init=10) to reduce sensitivity to initial centroid placement, and a fixed random state=42 for reproducibility.

6.2 Hierarchical Clustering (Validation)

Agglomerative Hierarchical Clustering with Ward linkage was applied independently to the same scaled feature matrix to validate the structure discovered by K-Means.

Advantages:

- Reveals nested cluster relationships not visible in a flat partitioning method
- Does not require a pre-specified number of clusters during the merge process
- Provides an independent structural check on K-Means results

Validation Metric	Value
K-Means Silhouette Score (k=4)	0.178
Hierarchical Silhouette Score (k=4)	0.175
Adjusted Rand Index (agreement)	0.683

The Adjusted Rand Index of 0.683 indicates substantial agreement between the two independent clustering methods, providing confidence that the four-segment structure reflects genuine patterns in buyer behavior rather than an artifact of a single algorithm.

7. Optimal Cluster Selection

Two standard evaluation methods were used together to identify the statistically optimal number of clusters: the Elbow Method and Silhouette Analysis.

7.1 Elbow Method

The Elbow Method plots within-cluster sum of squares (inertia) against the number of clusters (k). The point where the rate of decrease sharply flattens indicates diminishing returns from adding further clusters.

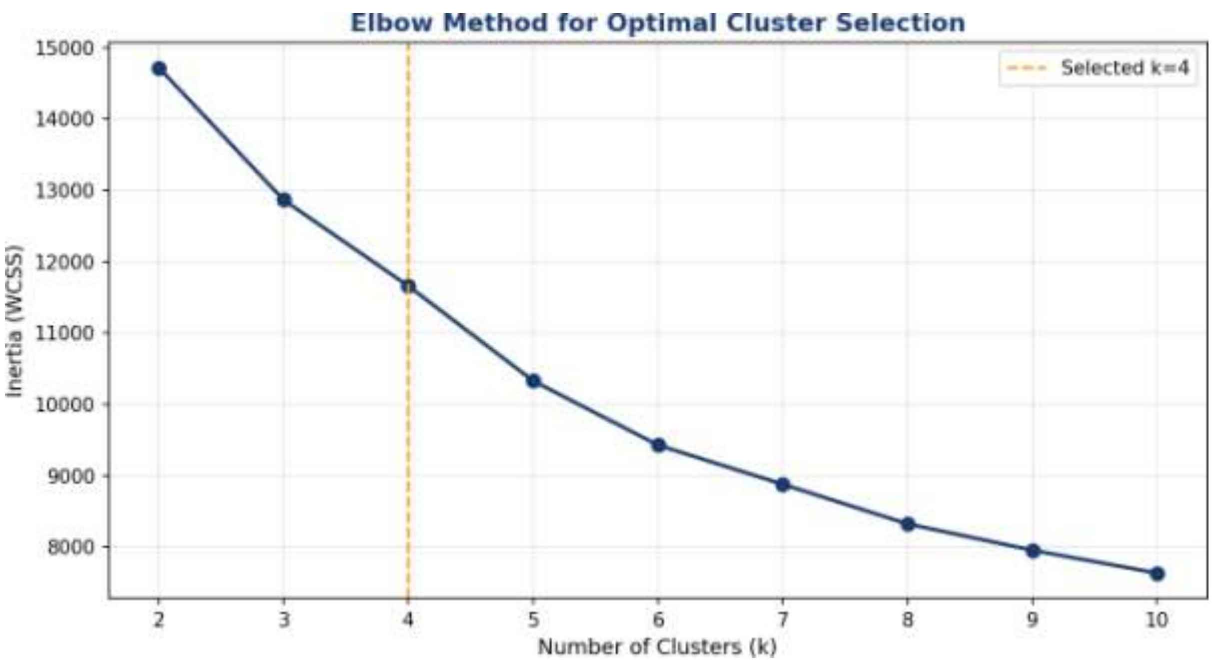

Figure 7.1 — Inertia vs. number of clusters. A visible bend appears around k=4, after which additional clusters yield progressively smaller reductions in inertia.

7.2 Silhouette Analysis

The Silhouette Score measures how well-separated and internally cohesive each cluster is, ranging from -1 (poor) to +1 (excellent). It was computed for k = 2 through k = 10.

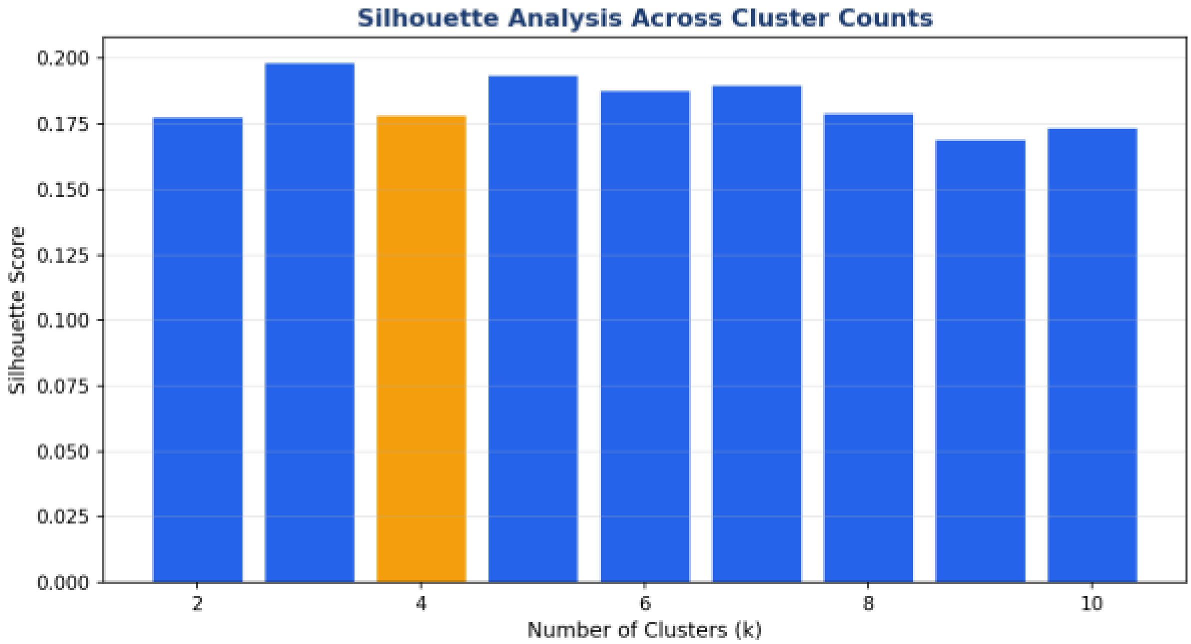

Figure 7.2 — Silhouette score across candidate cluster counts. k=4 was selected as the model to carry forward, balancing cluster quality with business interpretability and alignment with Parcl's four target buyer archetypes.

Model Selection Rationale: While marginally higher silhouette scores were observed at larger k values (e.g. k=8 or k=10), these produced fragmented, less interpretable segments with limited business actionability. k=4 was selected as it aligns with Parcl's four target buyer archetypes (Global Investors, First-Time Buyers, Corporate Buyers, Luxury Investors), is corroborated by the Elbow Method inflection point, and is independently validated by Hierarchical Clustering.

8. Cluster Interpretation & Buyer Segments

Each of the four clusters was profiled against investment purpose, geographic distribution, loan behavior, and customer demographics to translate statistical groupings into actionable buyer personas.

Figure 8.1 — Client count per identified buyer segment.

Segment	Buyer Type	Size	Key Characteristics
C1	Luxury Home Buyers	641 (32.1%)	Individual buyers, home-use focus, highest average unit price (~$41[illegible]K)
C2	Investment-Focused Buyers	454 (22.7%)	100% investment-purpose acquisitions, moderate loan usage
C3	First-Time / Value Buyers	802 (40.1%)	Individual, home-use, lowest average investment value
C4	Corporate Buyers	103 (5.1%)	100% company accounts, youngest average age, highest loan utilization

8.1 Cluster Statistical Profile

Metric	C1: Luxury	C2: Investment	C3: First-Time	C4: Corporate
Avg. Age	56.1	55.2	55.4	47.1
Avg. Satisfaction (1-5)	3.08	2.97	3.02	3.07
Avg. Units Purchased	3.83	3.55	3.56	3.70

Metric	C1: Luxury	C2: Investment	C3: First-Time	C4: Corporate
Avg. Total Investment	$1.57M	$1.13M	$1.08M	$1.26M
Avg. Unit Price	$418,885	$318,759	$306,032	$344,855
% Investment Purpose	19.5%	100.0%	0.0%	35.0%
% Loan Applied	35.1%	38.3%	36.7%	41.8%

8.2 Investment Value by Segment

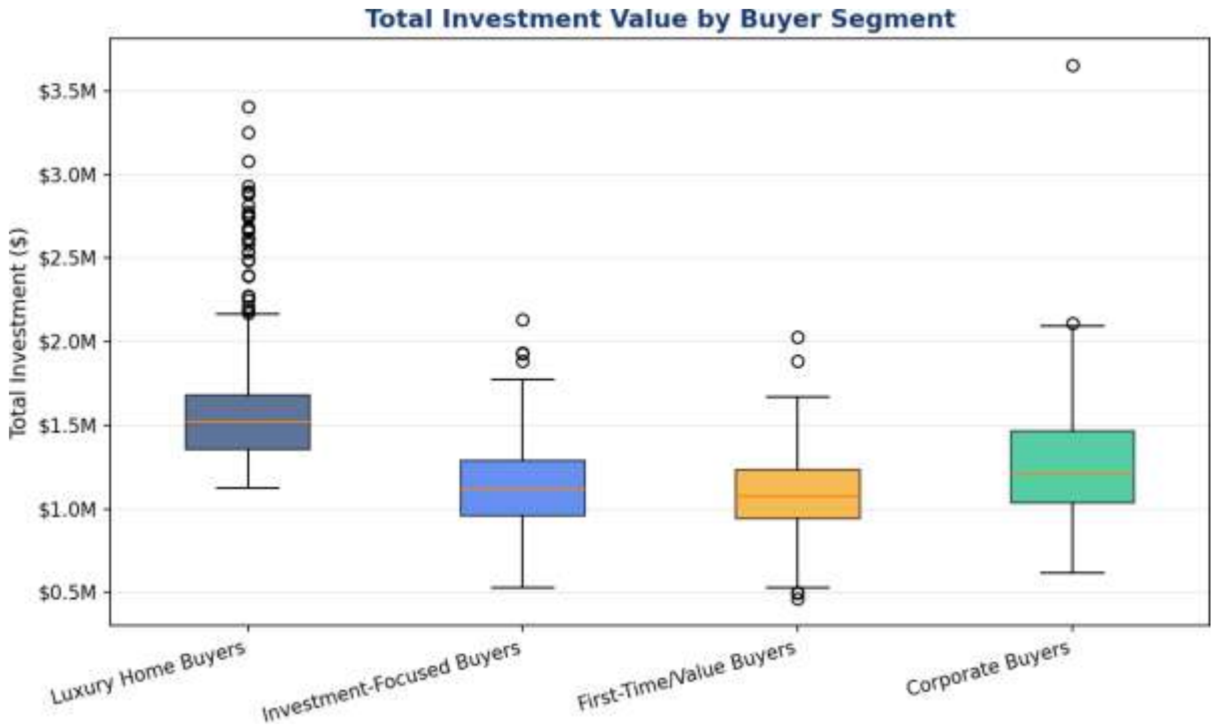

Figure 8.2 — Distribution of total client investment value by segment. Luxury Home Buyers show the highest median and widest spread, reflecting high-value, low-frequency purchases.

8.3 Acquisition Purpose & Financing by Segment

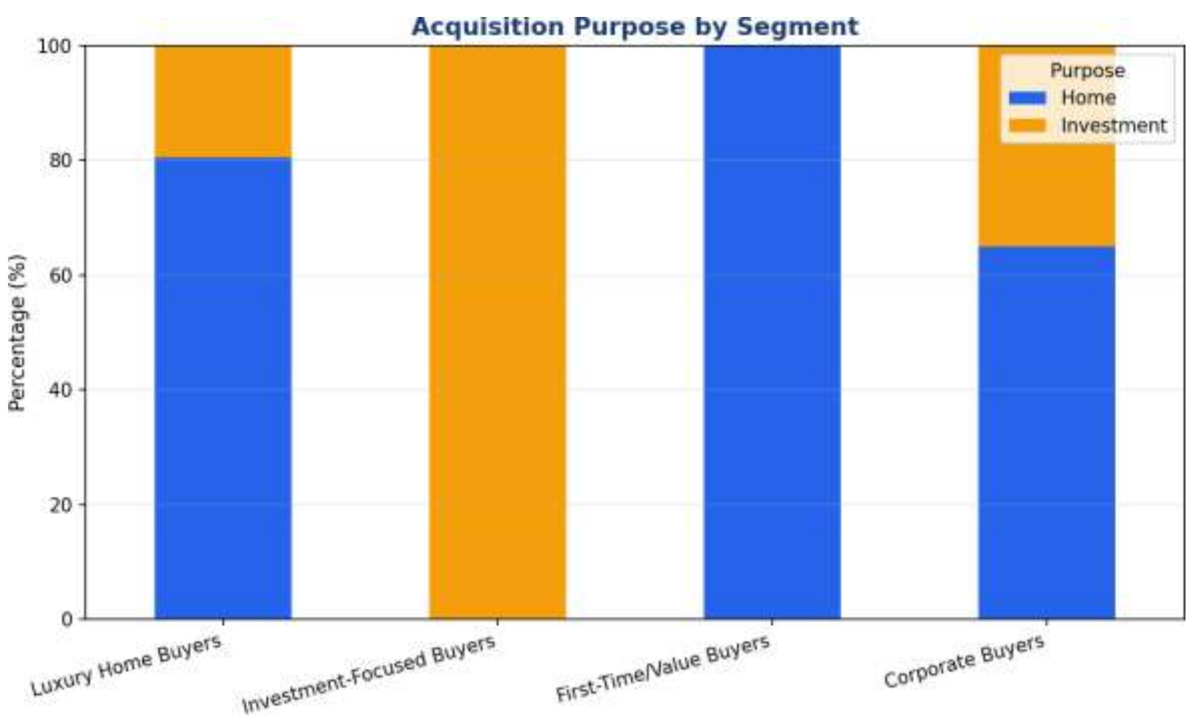

Figure 8.3 — Acquisition purpose composition confirms the defining feature of each segment, most notably the 100% investment orientation of C2.

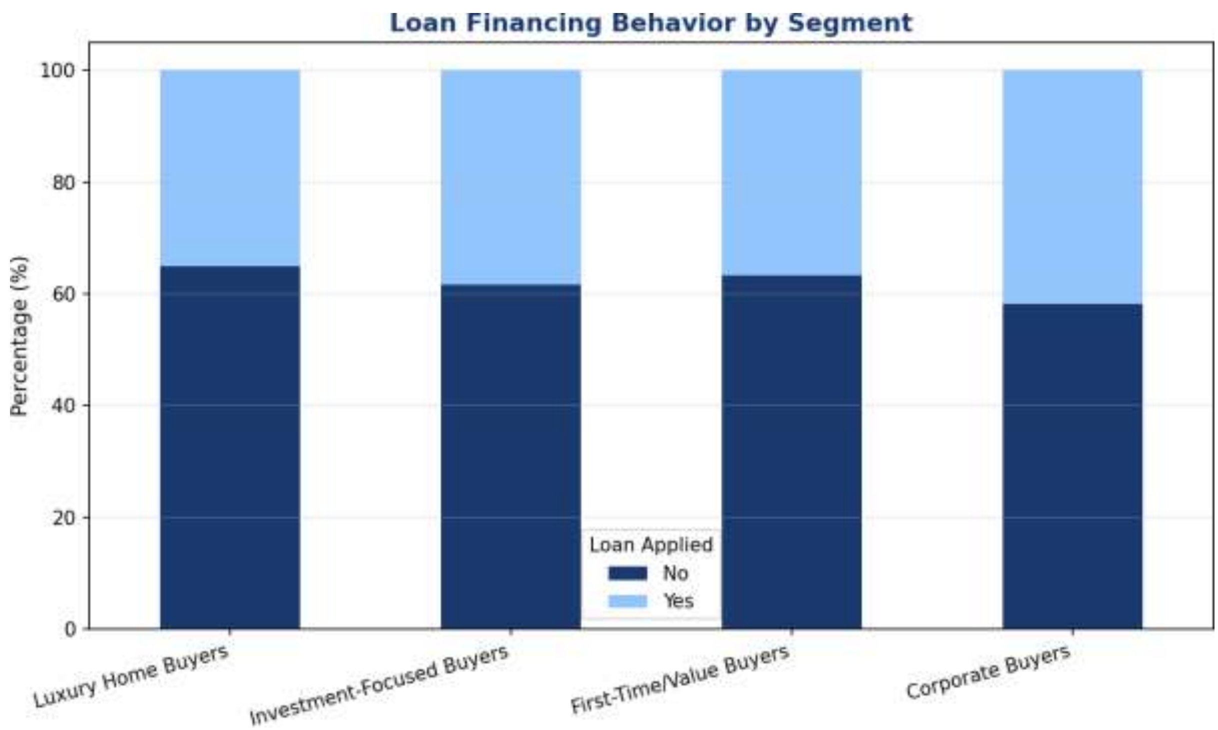

Figure 8.4 — Loan financing behavior by segment. Corporate Buyers show the highest reliance on financing.

9. Segment Deep-Dive Profiles

C1 — Luxury Home Buyers

32.1% of client base • 641 clients

Established individual buyers purchasing high-value properties primarily for personal/home use. This segment commands the highest average unit price across the portfolio and represents Parcl's premium residential customer base.

- Average age 56, skewing toward established, high-income households
- Average unit price of $418,885 — the highest of all segments
- Only 19.5% cite investment as their purpose, confirming a home-use orientation
- Moderate loan usage (35.1%) suggesting strong cash-purchase capacity

C2 — Investment-Focused Buyers

22.7% of client base • 454 clients

A purely investment-driven segment where 100% of transactions are classified as investment purpose. These buyers are strategically important for portfolio velocity and repeat engagement.

- 100% investment-purpose acquisitions — the defining trait of this segment
- Average total investment of $1.13M across an average of 3.55 units
- 38.3% loan utilization, the second-highest financing dependency
- Prime candidates for yield-focused and off-plan investment marketing

C3 — First-Time / Value Buyers

40.1% of client base • 802 clients

The largest segment by volume, composed of individual home-use buyers purchasing at the lowest average price point. This group represents Parcl's volume-driven residential base and entry-level market opportunity.

- Largest segment at 802 clients (40.1% of the total client base)
- Lowest average unit price ($306,032) and lowest average total investment ($1.08M)
- 0% investment-purpose acquisitions — exclusively home-use buyers
- 36.7% loan utilization, consistent with the portfolio average

C4 — Corporate Buyers

5.1% of client base • 103 clients

The smallest but structurally distinct segment, composed entirely of company accounts. This group shows the youngest average buyer age and the highest financing dependency, consistent with leveraged corporate acquisition strategies.

- 100% company-type accounts — the only segment defined by client_type
- Youngest average age at 47.1 years, reflecting active corporate portfolio managers
- Highest loan utilization at 41.8%, indicating leveraged acquisition strategies
- Mixed acquisition purpose (35% investment) reflecting both operational and portfolio buying

9.1 Satisfaction & Engagement by Segment

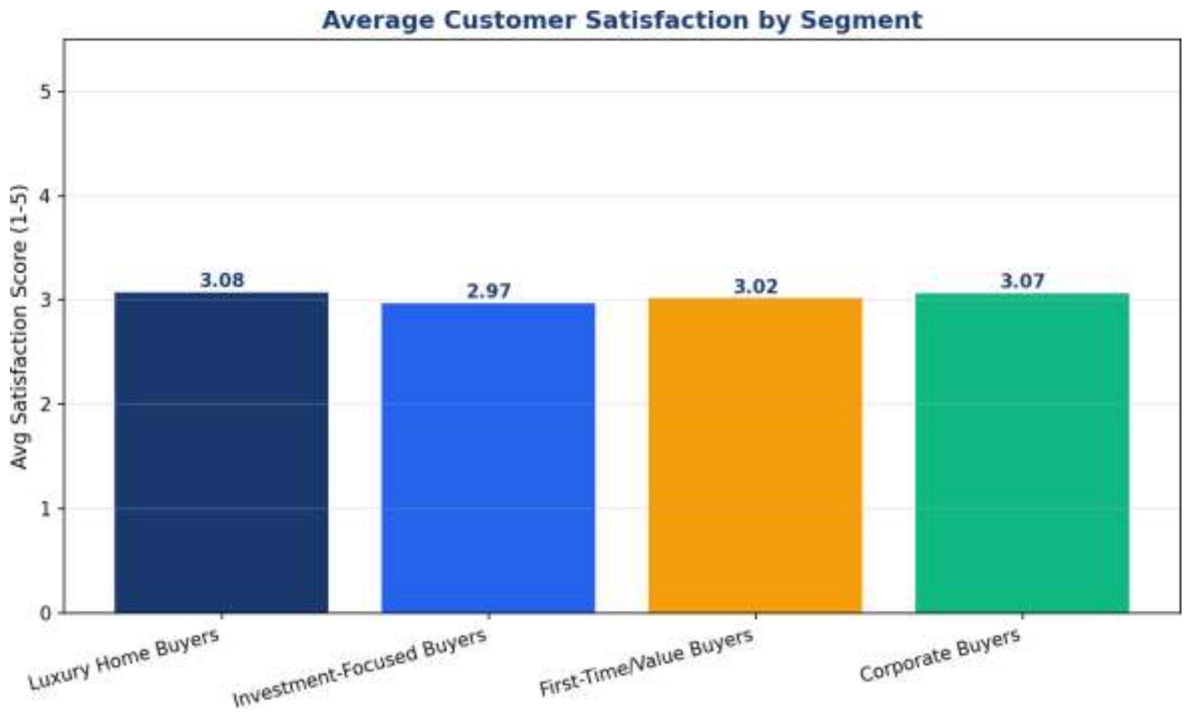

Figure 9.1 — Average customer satisfaction score by segment. Scores are broadly comparable across segments (2.97–3.08), indicating consistent service delivery regardless of buyer type.

10. Geographic & Behavioral Insights

Cross-referencing buyer segments with geography and referral channel data reveals where Parcl should focus regional marketing investment and which acquisition channels perform best for each buyer type.

Segment	Top Country	2nd	3rd
C1: Luxury Home Buyers	USA (497)	UK (37)	Canada (27)
C2: Investment-Focused	USA (349)	Canada (23)	UK (17)
C3: First-Time / Value	USA (612)	UK (36)	Canada (32)
C4: Corporate Buyers	USA (80)	France (5)	UK (5)

10.1 Referral Channel Effectiveness by Segment

Segment	Website	Agency	Client Referral
C1: Luxury Home Buyers	56.3%	35.3%	8.4%
C2: Investment-Focused	55.1%	34.4%	10.6%
C3: First-Time / Value	54.7%	35.5%	9.7%
C4: Corporate Buyers	51.5%	36.9%	11.7%

Website-driven acquisition is the leading channel across every segment, though Corporate Buyers show a comparatively higher reliance on Client referrals (11.7%) — consistent with B2B relationship-driven sales cycles.

10.2 Average Investment Value by Segment

Figure 10.1 — Average property unit price purchased by each buyer segment.

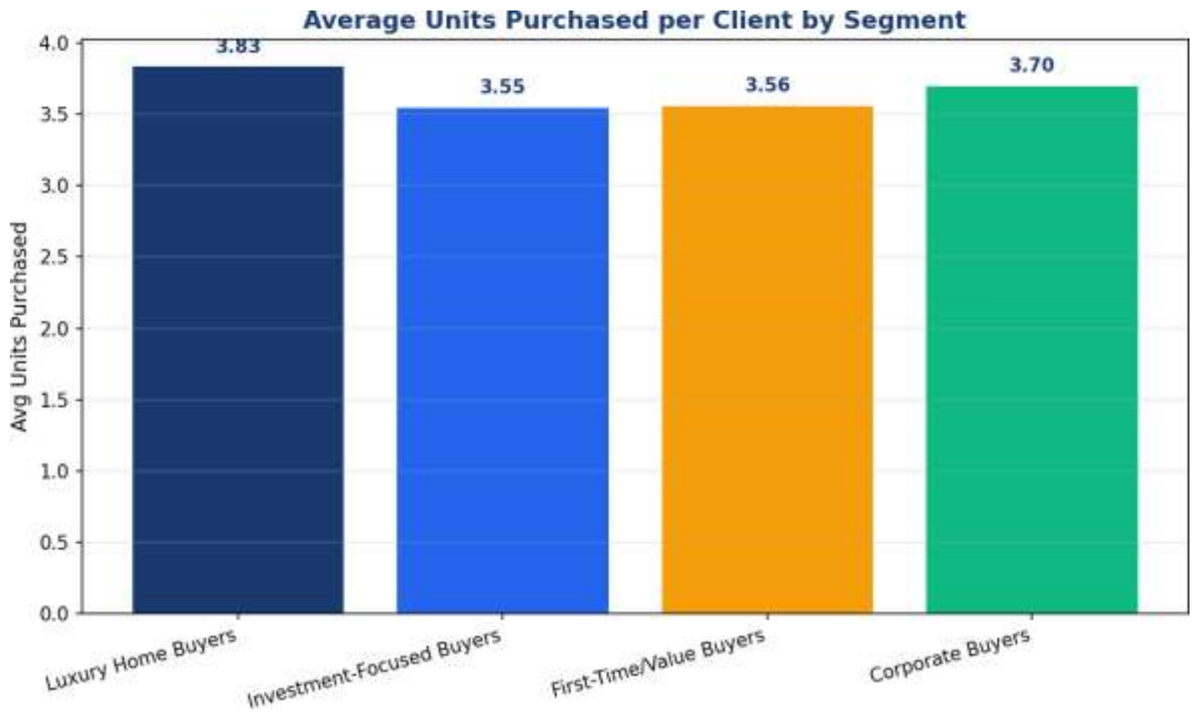

Figure 10.2 — Average number of units purchased per client by segment, indicating portfolio depth and repeat-purchase behavior.

10.3 Behavioral Summary

• The USA is the dominant market across every buyer segment, but international buyers (UK, Canada, Germany, France) are disproportionately represented in the Luxury and First-Time segments, suggesting cross-border interest in both premium and entry-level assets.

• Corporate buyers, while geographically concentrated in the USA, show the most diverse secondary country footprint (France, UK), pointing to international corporate portfolio strategies.

• Loan dependency rises with buyer youth and corporate structure, reinforcing the case for tailored financing products for the Corporate and Investment-Focused segments.

11. Streamlit Web Application

To operationalize these findings, an interactive Streamlit dashboard was designed to give Parcl's marketing and investment teams live, self-service access to the segmentation model outputs.

11.1 Dashboard Modules

Module	Function
Buyer Segmentation Overview	Displays overall cluster distribution and segment sizes
Investor Behavior Dashboard	Shows investment patterns, average deal size, and financing behavior by cluster
Geographic Buyer Analysis	Maps buyer segments by country and region
Segment Insights Panel	Provides descriptive statistics and persona summaries per cluster

11.2 User Controls

The dashboard allows users to dynamically filter all views by:

- Country
- Region
- Acquisition purpose (Home / Investment)
- Client type (Individual / Company)

11.3 Technical Architecture

The application is built on Python using Streamlit for the interface layer, pandas for data manipulation, and the pre-trained scikit-learn K-Means model (persisted via joblib/pickle) for real-time cluster assignment of new or filtered client records. Visualizations are rendered using Plotly and Matplotlib for interactive exploration.

12. Business Recommendations

1. Segment-Specific Marketing Campaigns

Design distinct marketing messaging for each of the four segments — premium positioning for Luxury Home Buyers, yield/ROI-focused messaging for Investment-Focused Buyers, affordability and first-time-buyer incentives for First-Time/Value Buyers, and portfolio/bulk-purchase offers for Corporate Buyers.

2. Tailored Financing Products

Given the elevated loan dependency observed in the Corporate (41.8%) and Investment-Focused (38.3%) segments, Parcl should explore partnership financing products, flexible mortgage structures, or bulk-acquisition credit lines targeted at these groups.

3. Channel Investment Optimization

Website remains the dominant acquisition channel across all segments; continued investment in digital acquisition (SEO, paid search, retargeting) is warranted, while Agency partnerships should be strengthened specifically for the Luxury and First-Time segments where they show above-average share.

4. Geographic Expansion Strategy

With international buyers concentrated in the UK, Canada, Germany, and France, Parcl should evaluate targeted international marketing and localized property showcases in these markets, particularly for the Luxury Home Buyer and Corporate segments.

5. Portfolio Recommendation Engine

Integrate the cluster model into Parcl's CRM to automatically recommend properties matching a client's segment profile (e.g. higher-value listings to Luxury Home Buyers, multi-unit packages to Corporate Buyers), improving conversion and reducing time-to-close.

6. Continuous Model Monitoring

Retrain the clustering model on a quarterly basis as new client and transaction data accumulates, using the Streamlit dashboard to track segment drift and emerging buyer archetypes over time.

13. Conclusion

This project introduces AI-driven buyer intelligence into the Parcl real estate platform. By applying K-Means clustering — validated through Hierarchical Clustering, the Elbow Method, and Silhouette Analysis — on a merged dataset of 2,000 clients and 10,000 property transactions, the analysis identified four statistically robust and commercially actionable buyer segments: Luxury Home Buyers, Investment-Focused Buyers, First-Time/Value Buyers, and Corporate Buyers.

These segments reveal investment behaviors, financing patterns, and geographic tendencies that traditional descriptive analytics could not detect on their own. By operationalizing these insights through the accompanying Streamlit dashboard, Parcl gains a continuously updatable, data-driven foundation for smarter marketing strategy, targeted property recommendations, and more precise investor engagement.

The methodology and infrastructure established here — from data cleaning through cluster validation — also provide a reusable framework that can be extended as Parcl's client base and transaction history continue to grow.

Deliverables

- Research paper covering exploratory data analysis, clustering methodology, and business insights (this document)
- Interactive Streamlit dashboard for live buyer segmentation analytics

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Prepared for Parcl Co. Limited in collaboration with Unified Mentor — July 2026